

$$\frac{1}{2} h_{\beta}^2 \partial^{\mu} h_{\alpha}^{\beta} \partial_{\mu} h_{\alpha}^{\gamma}$$

$$= \frac{1}{2} E_2 E_4 \left[q (K_4 - K_2) \right] \text{Tr } \overbrace{P}^{(-\eta^{\mu\nu})} \otimes (2 K_{1\mu} K_{3\nu})$$

$$= -E_2 E_4 (K_1 \cdot K_3) (-2 K_2 K_4) \approx + 2 (\bar{E} \omega)^2 (E_2 E_4) (1 - n_2 \cdot n_4)$$

$$- \frac{1}{2} h_{\beta}^2 \partial_{\alpha} h^{\mu\nu} \partial^{\beta} h_{\mu\nu}$$

$$= -\frac{1}{2} \left\{ \left[E_2 E_4 (2 K_{2\alpha} K_{4\beta} P^{\alpha\beta, \mu\nu}) \otimes (2 K_{1\mu} K_{3\nu}) \right] + 2 \left[(K_4 E_2 q) (E_{4\alpha\beta} P^{\alpha\beta, \mu\nu}) (2 K_{1\mu} K_{3\nu}) - (2 \leftrightarrow 4) \right] \right\}$$

$$= -\frac{1}{2} \left\{ 4 \left[E_2 E_4 \left(\frac{1}{2} (K_1 \cdot K_2) (K_3 \cdot K_4) + \frac{1}{2} (K_1 K_4) (K_3 K_2) - \frac{1}{2} (K_2 K_4) (K_1 K_3) \right) + 4 \left[(K_4 E_2 q) \left(2 \cdot \frac{1}{2} K_1 E_4 K_3 - E_{4\mu} K_{\mu} K_3 \right) - (K_2 E_4 q) (K_1 E_4 K_3) \right] \right] \right\}$$

$$\approx -\frac{(\bar{E} \omega)^2}{2} \left\{ 4 \left[E_2 E_4 \left(\frac{1}{2} (1 - n_2 v)(1 - n_4 v) + \frac{1}{2} (1 - n_4 v)(1 - n_2 v) - \frac{1}{2} (1 - n_2 n_4) \right) + 4 \left[(K_4 E_2 K_4) (\mathcal{O}(v^2)) + \mathcal{O}(v^2) \right] \right] \right\}$$

$$\approx -2 (\bar{E} \omega)^2 \left\{ (E_2 E_4) \left(\frac{1}{2} - n_2 v - n_4 v + \frac{n_2 n_4}{2} \right) \right\}$$

$$\approx -(\bar{E} \omega)^2 (E_2 E_4) \left[1 - 2 n v_2 - 2 n v_4 + n_2 n_4 \right]$$

$$-h_{\beta}^{\alpha} \partial_{\mu} h_{\alpha}^{\nu} \partial^{\mu} h_{\nu}^{\rho}$$

$$= - \left\{ \left[2 E_2^{\alpha\alpha} E_4^{\beta\beta} (K_2 K_4) P_{\alpha\beta}^{\rho\sigma} (2K_{1\rho} K_{3\sigma}) \right] \right. \\ \left. + \left[E_2^{\alpha\beta} E_4^{\beta\alpha} (K_4 q) P_{\alpha\beta}^{\rho\sigma} (2K_{1\rho} K_{3\sigma}) - (2 \leftrightarrow 4) \right] \right. \\ \left. + \left[E_2^{\alpha\beta} E_4^{\alpha\beta} (K_4 q) P_{\rho\sigma}^{\nu\beta} (2K_{1\rho} K_{3\sigma}) - (2 \leftrightarrow 4) \right] \right\}$$

$$= - \left\{ 4 \left[\frac{1}{2} (K_1 E_2 E_4 K_3) + \frac{1}{2} (K_3 E_2 E_4 K_1) - \frac{1}{2} (E_2 E_4) (K_1 K_3) \right] (K_2 K_4) \right. \\ \left. + 2 \left[\frac{1}{2} K_1 E_2 E_4 K_3 + \frac{1}{2} K_3 E_2 E_4 K_1 - \frac{1}{2} (E_2 E_4) (K_1 K_3) \right] (K_4 q) - 2 [2 \leftrightarrow 4] (-) \right. \\ \left. + 2 \left[\frac{1}{2} K_1 E_2 E_4 K_3 + \dots - \frac{1}{2} \dots \right] (1) - 2 \dots \right\}$$

$$\simeq - \left\{ -2 (E_2 E_4) (1 - \eta_2 \eta_4) + 2 [(E_2 E_4) (+K_2 K_4) + (E_2 E_4) (K_4 K_2)] \right\}$$

$$\simeq -2(E\omega)^2 (E_2 E_4) (1 - \eta_2 \eta_4) \quad \frac{1}{4} (1 - \eta_2 \eta_4) \} (E\omega)^2$$

$$\frac{1}{4} h^{\alpha\beta} h_{\alpha\mu} \partial_{\beta} h^{\mu\nu}$$

$$= \frac{1}{4} \left\{ 2 (K_2 K_4) (E_2 E_4) (-\eta^{\mu\nu}) (2K_{1\mu} K_{3\nu}) \right\} \simeq - (E\omega)^2 (E_2 E_4) (1 - \eta_2 \eta_4)$$

$$h_{\mu}^{\beta} \partial_{\alpha} h_{\beta}^{\mu}$$

$$= \left\{ E_2^{\alpha\beta} E_4^{\alpha\beta} K_{2\alpha} K_{4\beta} + \dots \right\}$$

$$h^\beta_\mu \partial_\nu h^\alpha_\beta \partial^\mu h^\nu_\alpha$$

$$= \left\{ \left[k_{2\nu} k_{4\mu} \epsilon_{2\beta}^\alpha \epsilon_{4\gamma}^\nu \right. \right. P_{\rho\sigma}^{\mu\beta} (2k_{1\rho} k_{3\sigma}) + (2 \leftrightarrow 4) \left. \right] \\ - \left[+ \epsilon_{2\mu\beta} \epsilon_{4\alpha}^\nu k_{2\nu} k_{4\mu} P_{\rho\sigma}^{\beta 2} + (2 \leftrightarrow 4) \right] (2k_{1\rho} k_{3\sigma}) \\ + \left[\epsilon_{2\mu}^\beta \epsilon_{4\rho\alpha} k_{4\nu} q^\mu P_{\rho\sigma}^{\alpha\nu} - (2 \leftrightarrow 4) \right] (2k_{1\rho} k_{3\sigma}) \left. \right\}$$

$$= \left\{ \left[(k_4 k_1) (k_3 \epsilon_2 \epsilon_4 k_2) + (k_4 k_3) (k_1 \epsilon_2 \epsilon_4 k_2) - (k_1 k_3) (k_4 \epsilon_2 \epsilon_4 k_2) \right. \right. \\ + (k_2 k_1) (k_3 \epsilon_4 \epsilon_2 k_4) + (k_2 k_3) (k_1 \epsilon_4 \epsilon_2 k_4) - (k_1 k_3) (k_2 \epsilon_4 \epsilon_2 k_4) \\ \left. \left. + 2 \left[(k_1 \epsilon_4 \epsilon_2 k_4) (k_4 k_3) + (k_3 \epsilon_2 \epsilon_4 k_2) (k_2 k_1) \right] \right] \right\} \epsilon_{2\alpha}^\beta \epsilon_{4\gamma}^\alpha$$

$$\simeq -2 \left[(k_1 \epsilon_2 k_4) (k_3 \epsilon_4 k_2) + (k_3 \epsilon_2 k_4) (k_1 \epsilon_4 k_2) - (k_4 \epsilon_2 k_2) (k_1 k_3) \right]$$

$$\simeq \left\{ \begin{aligned} & + \cancel{15 \epsilon_{11} \epsilon_{22} n_4} + \cancel{15 \epsilon_{11} n_4 n_4} - \cancel{15 \epsilon_{11} \epsilon_{22} n_4} \\ & 15 \epsilon_{22} \epsilon_4 k_2 + 15 \epsilon_{22} \epsilon_4 k_2 - \cancel{n_4 \epsilon_{22} \epsilon_4 k_2} + 15 \epsilon_{44} \epsilon_2 n_4 + 15 \epsilon_{44} \epsilon_2 n_4 \\ & + 2 \epsilon_{22} \epsilon_4 n_4 + 15 \epsilon_{22} \epsilon_4 n_2 \\ & - \cancel{n_2 \epsilon_{44} \epsilon_2 n_4} + 2 \cancel{15 \epsilon_{44} \epsilon_2 n_4} + 2 \cancel{15 \epsilon_{22} \epsilon_4 n_2} \end{aligned} \right\}$$

$$\simeq 4 \neq 2 \cdot \cancel{n_4 \epsilon_{22} \epsilon_4 n_2} \} \epsilon_{2\alpha}^\beta \epsilon_{4\gamma}^\alpha n_2$$

$$\simeq (E\omega)^2 \left\{ 4 \cancel{15 \epsilon_{22} \epsilon_4 n_2} + 4 \cancel{15 \epsilon_{44} \epsilon_2 n_4} \right\}$$